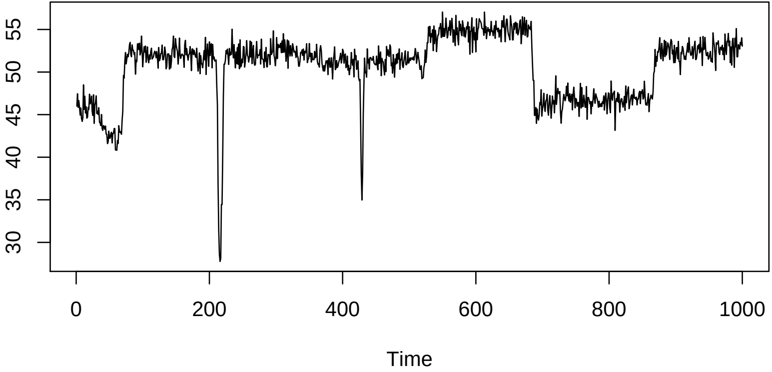
Influence in the world of changepoints

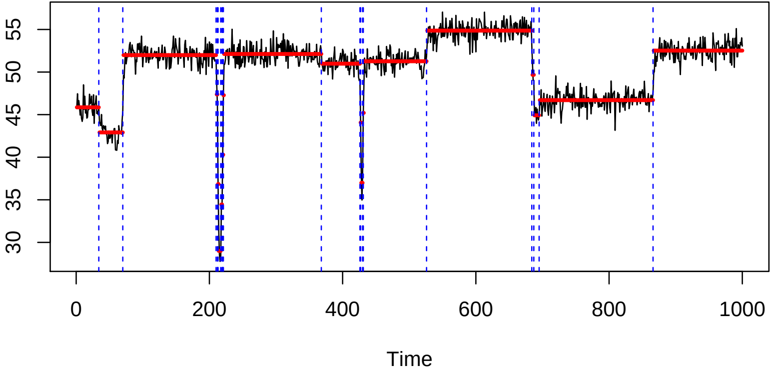
Rebecca Killick(r.killick@lancs.ac.uk)
Ines Wilms (Maastricht), David Matteson (Cornell)
Bernoulli-ims 2024

Contents

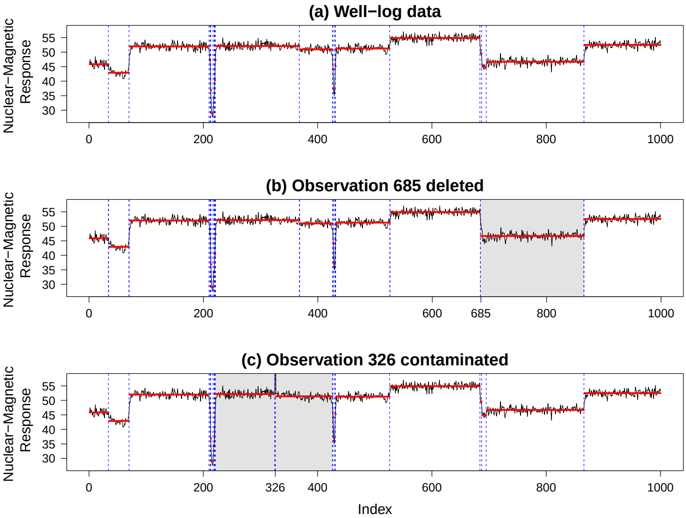
Motivation



Motivation



Motivation


(a) Well−log data
Nuclear−Magnetic Response
55
50
45
40
35
30
0
200
400
600
800
1000
(b) Observation 685 deleted
Nuclear−Magnetic Response
55
50
45
40
35
30
0
200
400
600
685
800
1000
(c) Observation 326 contaminated
Nuclear−Magnetic Response
55
50
45
40
35
30
0
200
326
400
600
800
1000
Index

Natural Questions

- Which data points are **influential** for obtaining the segmentation?
  - Changepoints versus Outliers
  - How to measure influence?
- How **stable** is the obtained segmentation?

Influence for Cpts

Sources of Inspiration:

- Regression Analysis: Measures of Influence (e.g., Cook's distance)
- Robust Statistics: Influence Functions

Two routes:

- Modifying an observation
- Leaving out an observation

Influence: Modify

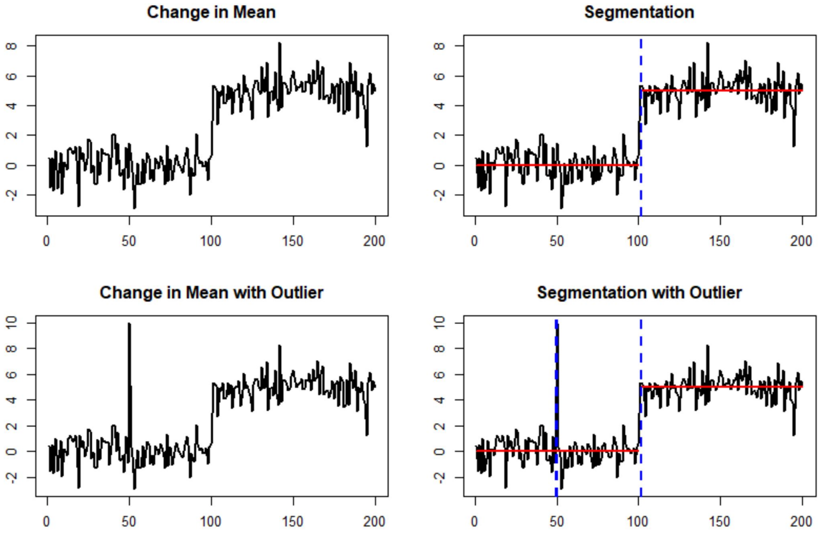
Roll through all observations $i = 1, \ldots, n$ and repeat:

- **Modify** observation i (Make an outlier)
- Obtain the **Expected Outlier Segmentation** (from theory)
- Obtain the **Actual Outlier Segmentation** (from preferred changepoint algorithm)

Outcomes:

- Changepoint remains in all segmentations
- Changepoint is deleted in some segmentations
- A new location is a changepoint in some segmentations

Influence: Modify

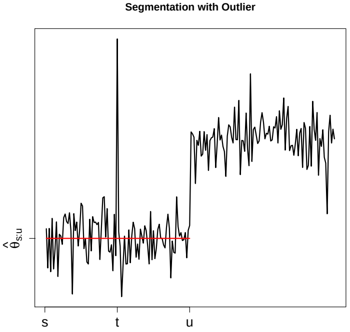


Influence: Modify

School of Mathematical Sciences | Lancaster University

Let

- y_t be an **outlier**
- $\boldsymbol{y}_{s:u}$ be segment containing the outlier.



Two extra changepoints at $t-1$ an t lead to **a reduction in cost** if

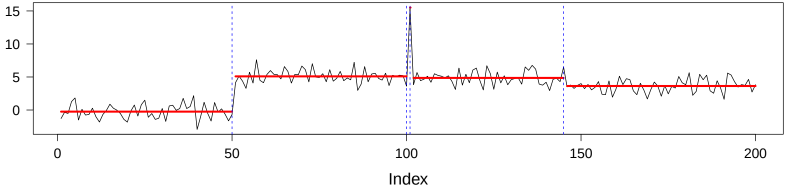
$$2\beta < \gamma(y_t, \hat{\theta}_{s:u})$$

where $\hat{\theta}_{s:u} = \underset{\theta}{\operatorname{argmin}} \sum_{j=s}^{u} \gamma(y_j; \theta)$ and $\gamma(\cdot)$ is the cost.

Influence: Modify

Running example:

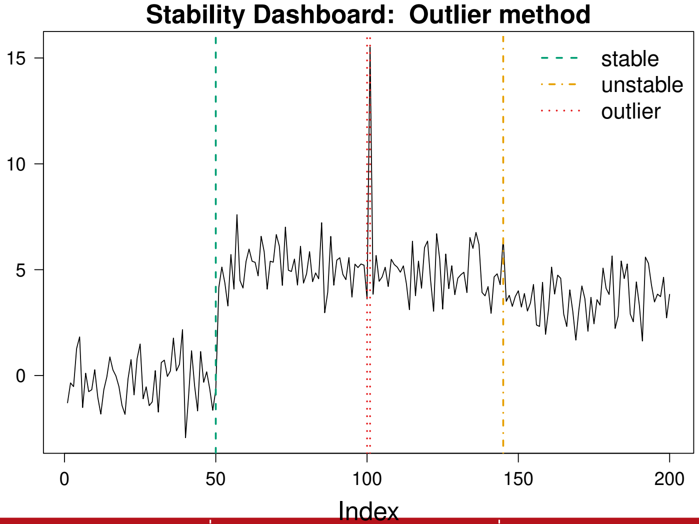
- One large change
- One outlier
- One small change




```
inf=influence(model,method="outlier")
```

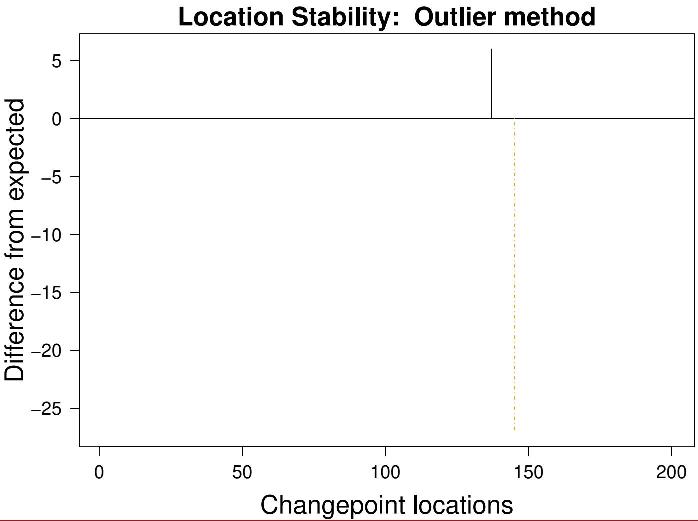
Influence: Modify


```
StabilityOverview(inf)
```



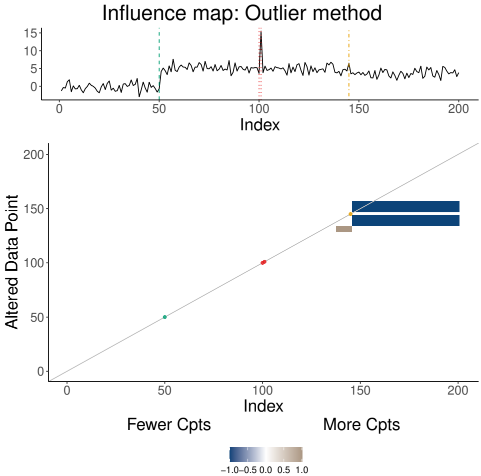
Influence: Modify


```
LocationStability(inf, type="Difference")
```

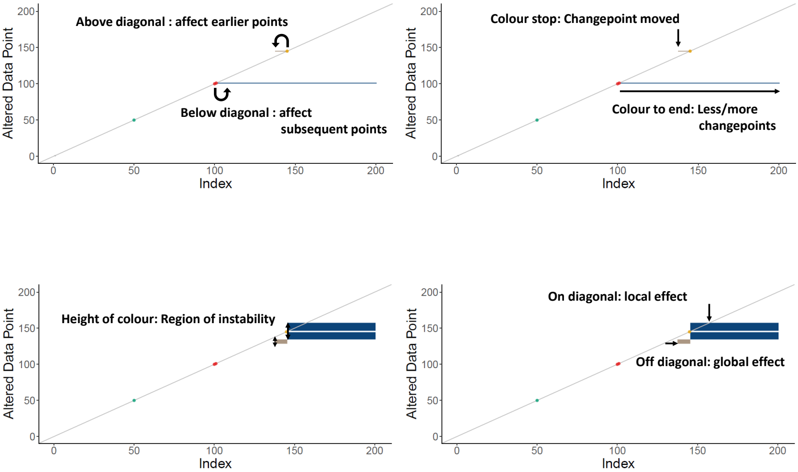


Influence: Modify

`InfluenceMap(inf,include.data=T)`

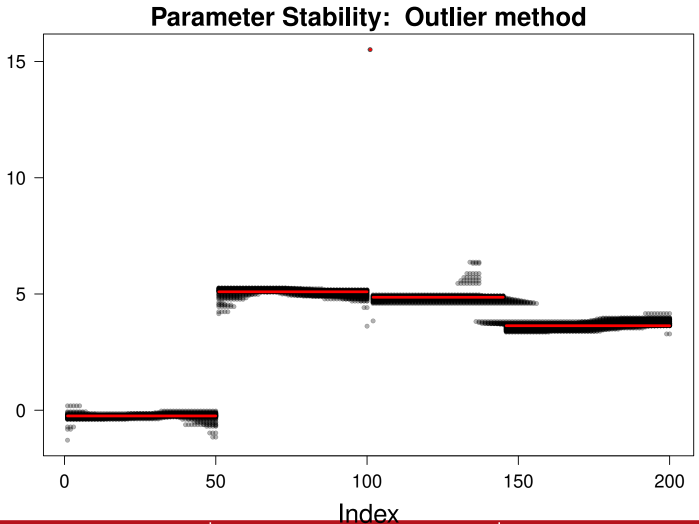


Influence: Map



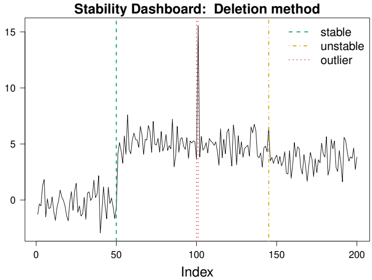
Influence: Modify

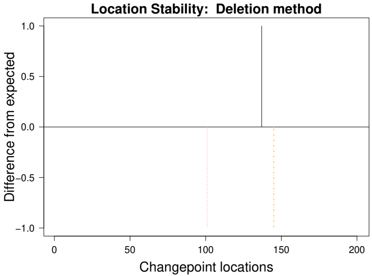
`ParameterStability(inf)`



Influence: Deletion

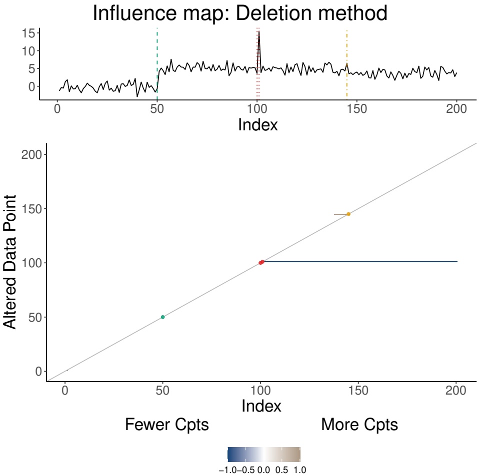

```
infdel=influence(model,method="delete")

LocationStability(infdel, type="Difference")

StabilityOverview(infdel)
```





Influence: Deletion

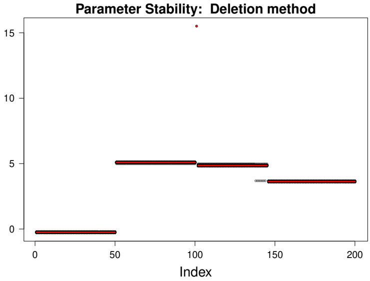

```
InfluenceMap(infdel,include.data=T)
```

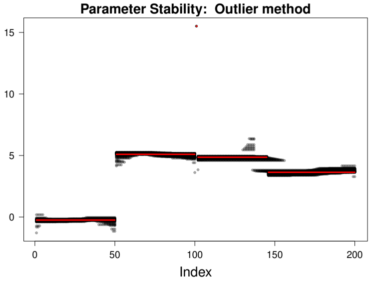


Influence: Deletion

`ParameterStability(inf)`

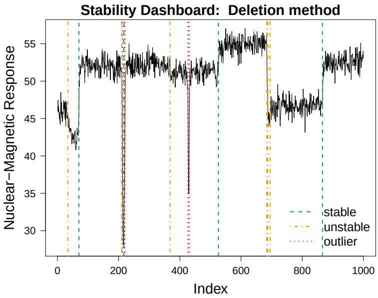
`ParameterStability(infdel)`

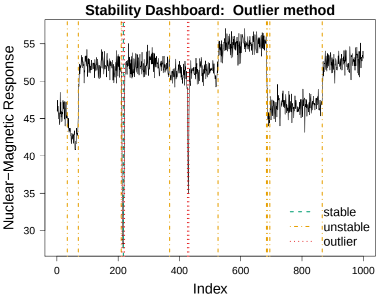




Application: Well-log

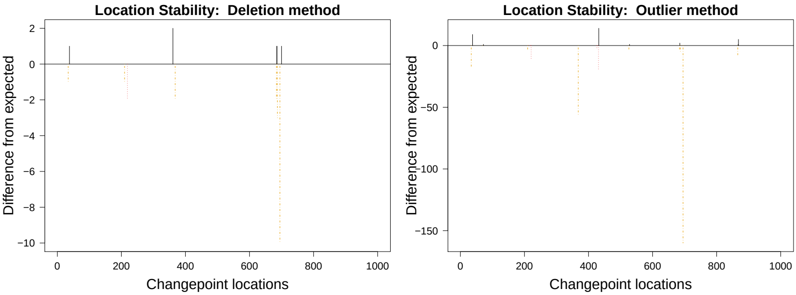

```
infwell=influence(model)

StabilityOverview(infwell)
```





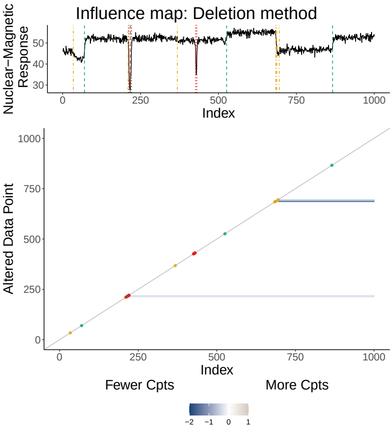
Application: Well-log

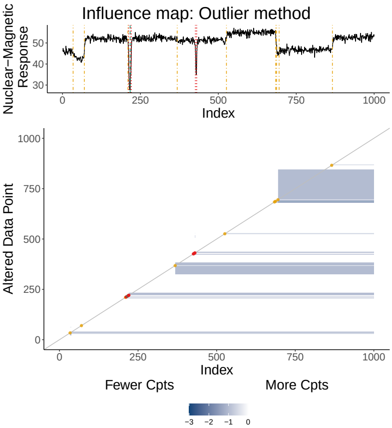

```
LocationStability(infwell, type="Difference")
```



Application: Well-log

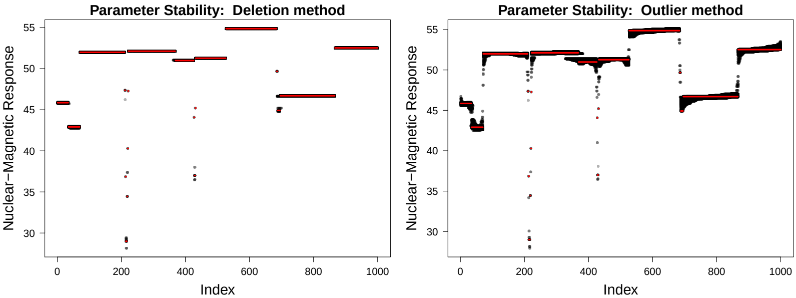
InfluenceMap(infwell,include.data=T)





Application: Well-log


```
ParameterStability(infwell)
```



`changepoint.influence`

Original Paper: Wilms, Killick, Matteson (2022) JCGS

New features (to be pushed to CRAN):

- `modify` - All, random, stratified
- `n.modify` - If not all, how many?
- `k` - Number of consecutive points to take for each of `n.modify`
- `same` - T/F, `k` outliers have the same mean?

Can work with any (univariate) changepoint method - user supplies `expected`.

Summary

- Indications of stability of changepoints is important for practitioners
- Decisions based on unstable changepoints may be flawed
- Having informative plots for assessing stability is important
- Modifying or deleting data points can give an indication of stability
- There are plenty more measures of stability we could use!

CRAN package: `changepoint.influence`